Amirreza Shojaei Baghini

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astro-arta.github.io GitHub LinkedIn

Research Interests

Extrasolar planet detection, planet formation, protoplanetary disks, and the development of open-source software for astronomical data analysis.

Education

M.Sc. in Physics 2025–Present
Sharif University of Technology, Tehran, Iran
Thesis (working title): *ISW in the Local Limit of Non-Local Gravity*
Advisor: Prof. Shant Baghram

B.Sc. in Mining Engineering 2019–2024
Amirkabir University of Technology, Tehran, Iran
Thesis: *Investigation of Zinc Recovery from Sulfide Ore*
Advisor: Dr. Haghighi

Relevant Coursework

Cosmology Sharif University of Technology
Topics: [Cosmic expansion, thermal history of cosmos, cosmological perturbation theory, structure formation, inflation].

Radio Astronomy Sharif University of Technology
Topics: [Radio astronomical fundamentals, emission mechanisms, radiative processes, thermal and non-thermal radio sources, spectral lines].

Astrobiology Sharif University of Technology
Topics: [Introductory planet formation, exoplanet detection methods, planetary atmospheres, emergence of life and evolution].

Astronomy Laboratory Sharif University of Technology
Topics: [observational techniques, instrumentation, data analysis and astronomical methods of data reduction].

Electrodynamics Sharif University of Technology
Topics: [Electrostatic, Maxwell equations, plane waves and propagation, waveguides, EM radiation, scattering and diffraction, relativity and EM fields].

Advanced Quantum Mechanics [Sharif University of Technology]
Topics: [Quantum Dynamics, angular momentum and spin, parity, perturbation theory, scattering, identical particles, relativistic quantum mehcanics].

Numerical Methods [Amirkabir University of Technology]
Topics: [Numerical integration, differential equations, numerical linear algebra, error analysis].

Research Experience

Visiting Student October 2026–Present
School of Astronomy, Institute for Research in Fundamental Sciences (IPM)

- Investigating Atacama Large Millimeter/ submillimeter Array (ALMA) observations, with a focus on radio spectroscopy.
- Developing a pipeline for the reduction and analysis of raw ALMA data.
- Attending the course *Physics of the Interstellar Medium*.

Intern June 2024–August 2024

NICICO, Sarcheshmeh Mining Complex – Leaching and Bioleaching Unit
Investigated the effect of pH on solvent extraction from copper pregnant leach solutions (PLS).

Technical Skills

Programming: Python
Scientific Libraries: NumPy, SciPy, Astropy, Astroquery, Matplotlib
Tools: Git, Linux, LaTeX, SQL

Awards & Honors

National Iranian Astronomy and Astrophysics Olympiad – Silver Medal 2017
National Iranian Astronomy and Astrophysics Olympiad – Bronze Medal 2016

Teaching & Outreach

Teaching Assistant – General Physics I 2025
Department of Physics, Sharif University of Technology
Led weekly problem-solving sessions and graded assignments for 43 undergraduate students.

Private Tutor – Physics and Astronomy 2017–Present
Independent
Tutored astronomy and physics for Olympiad participants, high-school students, and undergraduate students.

Languages

Persian: Native | **English:** Fluent | **French:** Intermediate | **Arabic:** Intermediate

Standardized Test Results

IELTS Academic: Overall Band 8.0
Listening: 8.0 | Reading: 8.5 | Writing: 7.5 | Speaking: 8.0